

ATAI Ultra-Safe Capsule Master

Documents Tree (60 Docs)

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Confidentiality: Confidential — For Tier 1 Investors, Strategic Partners, and
Technical Due Diligence Teams Only

Executive Overview

The ATAI Ultra-Safe Capsule program is a civil safety and industrial rescue concept for controlled high-mass descent, stabilized landing, and post-event survivability. The concept combines a rigid central core, 16 parachutes, 906 m fans, and 12 mechanical landing legs with dampers, supported by third-generation enabling technologies including non-Newtonian impact stiffening, maglev fan concepts using superconducting bearings, and pre-impact AI with LiDAR.

This master document tree is designed to demonstrate feasibility, safety, IP position, systems engineering discipline, verification readiness, and execution maturity without claiming certification, approval, or test completion. It is intended to support due diligence by showing that the program has a complete and traceable package of documents covering:

- Mission definition and use cases
- System architecture and subsystem interfaces
- Safety engineering and hazard controls
- Structural, aerodynamic, landing, and control feasibility
- Materials, thermal, power, and autonomy considerations
- Verification, validation, and test planning
- Manufacturing, supply chain, quality, and program execution
- IP, regulatory, ethics, and commercial readiness

The 60 documents below are grouped to mirror a real program lifecycle: concept definition, requirements, design basis, safety case, engineering analyses, verification planning, industrialization, and governance. Each item is intentionally framed as a planned or draft diligence artifact to show what evidence must exist before scale-up, partner engagement, or high-confidence investment decisions.

Master Documents Tree

Section A Program Definition and Governance

D001 Program Charter and Mission Statement

- Purpose: Defines the UltraSafe Capsule mission, civil safety scope, primary use cases, and program success criteria. Establishes the top-level intent for investors and partners.
- Owner: Program Director
- Status: Draft
- Evidence/Artifacts: Mission statement; scope boundaries; assumptions; non-goals; high-level timeline; stakeholder map; initial success metrics.

D002 Stakeholder Register and Governance Model

- Purpose: Identifies internal and external stakeholders, decision rights, escalation paths, and governance cadence. Supports disciplined oversight and partner alignment.
- Owner: Program Management Office (PMO)
- Status: Draft
- Evidence/Artifacts: Stakeholder matrix; RACI; steering committee structure; meeting cadence; escalation workflow; approval authorities.

D003 Program Business Case and Investment Thesis

- Purpose: Summarizes market need, value proposition, capital rationale, and strategic relevance for civil rescue and safety applications.
- Owner: Strategy Lead
- Status: Planned
- Evidence/Artifacts: Market sizing; demand drivers; revenue logic; investment scenarios; sensitivity analysis; strategic fit narrative.

D004 Program Master Schedule and Milestone Map

- Purpose: Establishes the integrated timeline from concept refinement through prototype, test, and partner validation phases.
- Owner: PMO Scheduler
- Status: Draft
- Evidence/Artifacts: Integrated Gantt; milestone gates; dependencies; critical path; resource loading; major risk dates.

D005 Configuration Management Plan

- Purpose: Defines how technical baselines, document revisions, interfaces, and change control are managed across the program.
- Owner: Configuration Manager
- Status: Draft
- Evidence/Artifacts: Baseline strategy; version control rules; change request process; configuration status accounting; document register.

D006 Risk Management Plan

- Purpose: Sets the framework for identifying, ranking, mitigating, and tracking program risks across technical, schedule, commercial, and supply-chain domains.
- Owner: Risk Manager

- Status: Draft
- Evidence/Artifacts: Risk matrix; thresholds; mitigation owners; review cadence; residual risk tracking; risk acceptance criteria.

D007 Ethics, Safety, and Responsible Innovation Policy

- Purpose: Defines the program's commitment to civil safety, responsible AI, transparent testing, and non-military use.
- Owner: Chief Compliance Officer
- Status: Draft
- Evidence/Artifacts: Ethical principles; AI governance rules; human safety commitments; prohibited use statement; escalation and review procedure.

D008 Systems Engineering Management Plan

- Purpose: Describes how requirements, architecture, interfaces, verification, and technical reviews are managed end-to-end.
 - Owner: Chief Systems Engineer
 - Status: Draft
 - Evidence/Artifacts: SE lifecycle; technical review gates; interface control process; requirements traceability method; review checklists.
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Section B Mission, Requirements, and Use Cases

D009 Mission Use Cases and Operational Scenarios

- Purpose: Defines the primary civil rescue and controlled descent scenarios the capsule is intended to support.
- Owner: Product Lead
- Status: Draft

- Evidence/Artifacts: Scenario descriptions; boundary conditions; mission envelopes; user profiles; operating assumptions.

D010 Top-Level Requirements Specification

- Purpose: Captures the top-level functional, performance, safety, and maintainability requirements for the capsule system.
- Owner: Requirements Manager
- Status: Draft
- Evidence/Artifacts: Requirement statements; rationale; verification method tags; priority ranking; trace links.

D011 ConOps: Concept of Operations

- Purpose: Describes how the capsule is prepared, deployed, guided, slowed, landed, inspected, and recovered in civil operations.
- Owner: Operations Lead
- Status: Draft
- Evidence/Artifacts: Operational flow; roles and responsibilities; state transitions; nominal and off-nominal sequences; recovery steps.

D012 System Boundary and Assumptions Document

- Purpose: Defines what is inside and outside the system boundary and records assumptions that affect architecture and verification.
- Owner: Chief Systems Engineer
- Status: Draft
- Evidence/Artifacts: Boundary diagram; external dependencies; environmental assumptions; regulatory assumptions; interface assumptions.

D013 Performance Requirements and Budget Allocation

- Purpose: Allocates performance targets across descent, stabilization, landing, survivability, and turnaround functions.
- Owner: Chief Systems Engineer
- Status: Draft
- Evidence/Artifacts: Budget tables; mass/power/load allocations; descent energy assumptions; landing tolerances; margin policy.

D014 Safety Objectives and Safety Integrity Targets

- Purpose: Defines the safety goals, target risk levels, and acceptance criteria guiding the safety case.
- Owner: Safety Lead
- Status: Draft
- Evidence/Artifacts: Safety objectives; hazard tolerance criteria; risk thresholds; safety integrity mapping; rationale.

D015 Environmental and Deployment Conditions Specification

- Purpose: Captures the environmental envelopes relevant to deployment, descent, landing, storage, transport, and maintenance.
 - Owner: Environmental Lead
 - Status: Draft
 - Evidence/Artifacts: Temperature/wind/dust/rain conditions; altitude bands; shock/vibration loads; storage constraints; usage limits.
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Section C System Architecture and Interface Control

D016 System Architecture Overview

- Purpose: Presents the end-to-end capsule architecture and major functional blocks.
- Owner: Lead Architect
- Status: Draft
- Evidence/Artifacts: Architecture diagrams; functional decomposition; block interactions; architecture rationale; key trade-offs.

D017 Rigid Central Core Design Basis

- Purpose: Defines the role, structural intent, load path, and protection functions of the rigid central core.
- Owner: Structural Lead
- Status: Draft
- Evidence/Artifacts: Core concept sketches; load path diagrams; protection zones; material assumptions; integration constraints.

D018 Descent and Stabilization Subsystem Architecture

- Purpose: Describes the combined parachute and fan-assisted descent architecture for controlled deceleration and attitude management.
- Owner: Aerodynamics Lead
- Status: Draft
- Evidence/Artifacts: Descent sequence; parachute/fan interactions; control states; stabilization logic; performance envelopes.

D019 Landing System Architecture

- Purpose: Defines the 12 mechanical landing legs with dampers and their role in impact absorption and post-touchdown stability.
- Owner: Landing Systems Lead
- Status: Draft
- Evidence/Artifacts: Leg layout; damper principles; load distribution; retraction/deployment concept; landing geometry.

D020 Power and Energy Architecture

- Purpose: Describes the electrical and energy architecture supporting sensing, control, actuation, fans, and safety functions.
- Owner: Power Systems Lead
- Status: Draft
- Evidence/Artifacts: Power tree; storage strategy; distribution topology; emergency power path; load profile estimates.

D021 Avionics and Embedded Computing Architecture

- Purpose: Defines compute, sensing, control, and fault-management electronics architecture.
- Owner: Avionics Lead
- Status: Draft
- Evidence/Artifacts: Computer architecture; bus topology; redundancy concept; fault isolation strategy; timing budgets.

D022 Interface Control Document Index

- Purpose: Lists and governs all major interface control documents across subsystems and external systems.
- Owner: Interface Manager
- Status: Draft

- Evidence/Artifacts: ICD register; interface ownership; status list; revision history; unresolved interface issues.

D023 Mass Properties and Layout Control Document

- Purpose: Tracks mass, center-of-gravity, envelope, and packaging assumptions for the whole system.
 - Owner: Systems Engineering Lead
 - Status: Draft
 - Evidence/Artifacts: Mass breakdown; CG envelope; layout drawings; change impact rules; margin reserves.
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Section D Safety Engineering and Hazard Control

D024 Preliminary Hazard Analysis (PHA)

- Purpose: Identifies major hazards associated with descent, landing, maintenance, transport, and human interaction.
- Owner: Safety Lead
- Status: Draft
- Evidence/Artifacts: Hazard list; severity/probability ranking; hazard causes; initial mitigations; residual risk notes.

D025 HARA: Hazard Analysis and Risk Assessment

- Purpose: Expands hazard analysis into structured risk assessment for operational and off-nominal states.
- Owner: Safety Lead
- Status: Draft

- Evidence/Artifacts: Hazard scenarios; risk matrix; controllability and exposure factors; mitigation mapping; acceptance rationale.

D026 FMEA: System-Level Failure Modes and Effects Analysis

- Purpose: Analyzes likely failure modes and their effects across the complete capsule system.
- Owner: Reliability Lead
- Status: Draft
- Evidence/Artifacts: Failure modes; effects; detection methods; severity/occurrence rankings; recommended actions.

D027 Fault Tree Analysis (FTA) Summary

- Purpose: Shows top-event pathways for critical safety events and identifies combinations of failures requiring control.
- Owner: Safety Analysis Lead
- Status: Draft
- Evidence/Artifacts: Fault trees; top events; minimal cut sets; logical dependencies; mitigation closure status.

D028 Safety Case Structure and Argument Plan

- Purpose: Defines the logical structure for proving the safety case through evidence, analysis, and test.
- Owner: Safety Case Manager
- Status: Draft
- Evidence/Artifacts: Safety claims; argument structure; evidence mapping; open points; assurance strategy.

D029 Emergency Modes and Abort Logic Specification

- Purpose: Defines safe-mode states, abort triggers, recovery states, and degraded-operation behavior.
- Owner: Controls Lead
- Status: Draft
- Evidence/Artifacts: Mode table; triggers; fallback actions; fail-safe logic; operator alerts; recovery conditions.

D030 Human Factors and Operational Safety Plan

- Purpose: Addresses safe human interaction during handling, deployment, monitoring, maintenance, and recovery.
 - Owner: Human Factors Lead
 - Status: Draft
 - Evidence/Artifacts: Operator tasks; ergonomic considerations; warning labels; training assumptions; interface usability notes.
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Section E Structural, Materials, and Mechanical Design

D031 Structural Design Basis Report

- Purpose: Establishes the loads, margins, constraints, and design philosophy for the capsule structure.
- Owner: Structural Lead
- Status: Draft
- Evidence/Artifacts: Load cases; margin policy; design allowables; materials assumptions; structural sizing rationale.

D032 20-Ton Capsule Structural Load Case Library

- Purpose: Defines the canonical load cases for launch, descent, dynamic impact, landing, handling, and transport.
- Owner: Structural Analysis Lead
- Status: Draft
- Evidence/Artifacts: Load tables; combinations; limit/ultimate definitions; worst-case scenarios; safety factors.

D033 Landing Leg and Damper Design Report

- Purpose: Details the 12-leg landing system, damping strategy, and load-sharing behavior for impact mitigation.
- Owner: Landing Systems Engineer
- Status: Draft
- Evidence/Artifacts: Leg geometry; damper curves; stroke limits; attachment points; load path verification; failure considerations.

D034 Fan Module Mechanical Design Report

- Purpose: Covers the 9×6 m fan concept, structural integration, mechanical constraints, and actuation support.
- Owner: Mechanical Lead
- Status: Draft
- Evidence/Artifacts: Fan envelope; mounting architecture; thrust/load assumptions; vibration considerations; maintainability notes.

D035 Parachute System Design Report

- Purpose: Defines the 16-parachute configuration, deployment logic, pack architecture, and interaction with vehicle dynamics.
- Owner: Aerodynamics/Recovery Lead
- Status: Draft
- Evidence/Artifacts: Parachute sizing logic; line load estimates; deployment sequence; reefing concept; packing constraints.

D036 Materials Selection and Compatibility Report

- Purpose: Justifies material choices for structure, damping, thermal interfaces, and environmental durability.
- Owner: Materials Lead
- Status: Draft
- Evidence/Artifacts: Candidate materials; property tables; corrosion/aging data; compatibility matrix; trade-off summary.

D037 Non-Newtonian Impact Stiffening Concept Note

- Purpose: Documents the third-generation impact mitigation concept using non-Newtonian fluids or media for energy absorption.
- Owner: Advanced Materials Lead
- Status: Draft
- Evidence/Artifacts: Material behavior summary; impact response curves; packaging concept; test hypotheses; safety considerations.

D038 Structural Dynamics and Vibration Control Report

- Purpose: Assesses modal behavior, resonance risks, and damping needs for structural integrity and sensor stability.
 - Owner: Dynamics Lead
 - Status: Draft
 - Evidence/Artifacts: Modal estimates; vibration sources; damping strategy; coupling analysis; test plan inputs.
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Section F Aerodynamics, Descent, and Landing Control

D039 Aerodynamic Descent Analysis

- Purpose: Evaluates drag, stability, attitude behavior, and interaction among parachutes, body, and fans.
- Owner: Aerodynamics Lead
- Status: Draft
- Evidence/Artifacts: CFD summaries; descent coefficients; stability curves; trajectory assumptions; sensitivity results.

D040 Parachute Deployment Dynamics Study

- Purpose: Analyzes deployment shock, sequencing, inflation dynamics, and safe activation thresholds for the 16-chute system.
- Owner: Recovery Systems Lead
- Status: Draft
- Evidence/Artifacts: Deployment sequence logic; shock estimates; timing diagrams; failure cases; deployment envelopes.

D041 Fan-Assisted Control and Maglev Bearing Concept Study

- Purpose: Explores the maglev fan concept using superconducting bearings for low-friction, high-durability rotation.
- Owner: Advanced Propulsion Lead
- Status: Draft
- Evidence/Artifacts: Concept layout; magnetic bearing principles; thermal assumptions; reliability considerations; test hypotheses.

D042 Pre-Impact AI and LiDAR Sensing Specification

- Purpose: Defines the sensing and AI functions that estimate terrain, descent state, and landing conditions before touchdown.
- Owner: Autonomy Lead
- Status: Draft
- Evidence/Artifacts: Sensor suite definition; LiDAR coverage; inference functions; latency budgets; safety fallbacks.

D043 Guidance, Navigation, and Control (GNC) Architecture

- Purpose: Describes how sensing and control inputs produce stable descent, correction, and landing alignment.
- Owner: GNC Lead
- Status: Draft
- Evidence/Artifacts: Control loops; state estimation; actuation map; mode transitions; performance margins.

D044 Landing Footprint and Terrain Interaction Study

- Purpose: Analyzes contact geometry, ground pressure, slope tolerance, and touchdown stability.
 - Owner: Landing Analysis Lead
 - Status: Draft
 - Evidence/Artifacts: Terrain models; footprint calculations; slope sensitivity; ground interaction assumptions; stability criteria.
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Section G Software, AI, and Cybersecurity

D045 Software Development Plan

- Purpose: Defines software lifecycle practices, coding standards, reviews, testing, and release management.
- Owner: Software Manager
- Status: Draft
- Evidence/Artifacts: Development lifecycle; coding standards; review gates; issue tracking; release criteria.

D046 AI Safety and Model Governance Plan

- Purpose: Governs the use of AI for pre-impact sensing, decision support, and anomaly detection in a safety-critical context.
- Owner: AI Safety Lead
- Status: Draft
- Evidence/Artifacts: Model scope; training data governance; validation criteria; override rules; human oversight design.

D047 Data Management and Sensor Calibration Plan

- Purpose: Specifies how sensor data, calibration records, logs, and test datasets are handled and preserved.
- Owner: Data Engineering Lead
- Status: Draft
- Evidence/Artifacts: Data schemas; calibration methods; retention policy; versioning; lineage records.

D048 Cybersecurity and Secure Boot Architecture

- Purpose: Addresses system integrity, authentication, update security, and protection against unauthorized access or tampering.
- Owner: Cybersecurity Lead

- Status: Draft
- Evidence/Artifacts: Threat model; secure boot design; encryption/key handling; update process; access control rules.

D049 Fault Detection, Isolation, and Recovery (FDIR) Specification

- Purpose: Defines how faults are detected, isolated, logged, and managed without compromising safety.
 - Owner: Controls/Software Lead
 - Status: Draft
 - Evidence/Artifacts: Fault detection logic; recovery state machines; fault logs; redundancy handling; safe-mode rules.
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Section H Verification, Validation, and Test Readiness

D050 Verification and Validation (V&V) Master Plan

- Purpose: Provides the overall strategy for proving requirements compliance through analysis, inspection, test, and demonstration.
- Owner: Verification Lead
- Status: Draft
- Evidence/Artifacts: Verification matrix; method selection; gate criteria; test hierarchy; traceability links.

D051 Requirements Traceability Matrix

- Purpose: Traces each top-level requirement to design outputs, analyses, inspections, and verification activities.
- Owner: Requirements Manager
- Status: Draft
- Evidence/Artifacts: Requirement IDs; linked evidence; verification status; coverage gaps; change traceability.

D052 Prototype Test Strategy and Development Test Plan

- Purpose: Defines the staged test program from component tests through integrated subsystem and system-level demonstrations.
- Owner: Test Director
- Status: Draft
- Evidence/Artifacts: Test phases; objectives; entry/exit criteria; instrumentation plan; test matrix.

D053 Component Qualification Test Specifications

- Purpose: Lists qualification-style test requirements for critical components such as legs, fans, sensors, and control electronics.
- Owner: Test Engineering Lead
- Status: Draft
- Evidence/Artifacts: Test procedures; environmental conditions; pass/fail criteria; sample counts; data capture requirements.

D054 Instrumentation and Telemetry Plan

- Purpose: Defines the measurement architecture for structural, aerodynamic, thermal, electrical, and control validation.
- Owner: Test Instrumentation Lead
- Status: Draft
- Evidence/Artifacts: Sensor list; channels; sampling rates; calibration approach; telemetry architecture; synchronization method.

D055 Ground Test Facility Requirements

- Purpose: Specifies the facilities needed for structural, systems, deployment, fan, and landing simulations.
- Owner: Test Facilities Manager
- Status: Draft

- Evidence/Artifacts: Facility requirements; power/rigging needs; safety systems; environmental controls; occupancy rules.

D056 Full-Scale Demonstration Test Readiness Review Package

- Purpose: Provides the readiness evidence required before any large-scale integrated demonstration.
 - Owner: Test Director
 - Status: Planned
 - Evidence/Artifacts: Readiness checklist; safety approvals; test objectives; emergency procedures; configuration baseline; open issues list.
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Section I Manufacturing, Supply Chain, and Quality

D057 Manufacturing Plan and Build Sequence

- Purpose: Describes how the capsule and subsystems will be fabricated, assembled, integrated, and checked.
- Owner: Manufacturing Lead
- Status: Draft
- Evidence/Artifacts: Build flow; tooling needs; assembly sequence; work instructions; inspection points.

D058 Supply Chain and Critical Vendor Strategy

- Purpose: Identifies critical components, sourcing risks, qualification needs, and single-point dependencies.
- Owner: Supply Chain Lead
- Status: Draft
- Evidence/Artifacts: Approved vendor list; lead times; alternates; qualification criteria; sourcing risk register.

D059 Quality Management Plan

- Purpose: Defines quality controls, inspection standards, nonconformance handling, and corrective action processes.
- Owner: Quality Manager
- Status: Draft
- Evidence/Artifacts: Quality policy; inspection plan; NCR workflow; CAPA process; audit schedule.

D060 Serialization, Traceability, and Maintenance Plan

- Purpose: Ensures each major item is serially tracked, maintainable, and recoverable through its lifecycle.
- Owner: Reliability and Logistics Lead
- Status: Draft
- Evidence/Artifacts: Serialization scheme; maintenance intervals; spare parts logic; life-limit tracking; service records.

Minimum Evidence Pack for Royal UAE (10 items)

The following ten items represent the minimum diligence package suitable for a high-level partner or government-facing review. These are not claims of approval; they are the essential artifacts expected to establish feasibility, safety intent, and execution discipline.

1. D001 Program Charter and Mission Statement
Confirms civil safety scope and strategic intent.
2. D010 Top-Level Requirements Specification
Shows what the system must do and how success is measured.

3. D011 ConOps: Concept of Operations
Explains how the capsule will be used in real-world rescue/safety scenarios.
 4. D016 System Architecture Overview
Demonstrates how the main subsystems fit together.
 5. D024 Preliminary Hazard Analysis (PHA)
Establishes the initial safety hazard baseline.
 6. D028 Safety Case Structure and Argument Plan
Shows how safety will be evidenced and argued.
 7. D031 Structural Design Basis Report
Provides the core structural engineering assumptions for the 20-ton capsule.
 8. D042 Pre-Impact AI and LiDAR Sensing Specification
Demonstrates the intended pre-touchdown sensing and decision-support approach.
 9. D050 Verification and Validation (V&V) Master Plan
Shows how the design will be proven through analysis and testing.
 10. D058 Supply Chain and Critical Vendor Strategy
Indicates execution readiness and industrialization realism.
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Appendix Definitions

- TRL (Technology Readiness Level): A scale used to estimate the maturity of a technology, from basic concept to proven operational system.
- FMEA (Failure Modes and Effects Analysis): A structured method to identify possible failures, their effects, and mitigations.
- HARA (Hazard Analysis and Risk Assessment): A systematic process for identifying hazards and evaluating associated risk.
- SIL (Safety Integrity Level): A measure of required risk reduction for safety-related functions, typically used in safety engineering contexts.

- ConOps (Concept of Operations): A description of how a system is intended to be used in practice.
 - FDIR (Fault Detection, Isolation, and Recovery): Methods for finding faults, isolating their sources, and recovering safely.
 - PHA (Preliminary Hazard Analysis): Early-stage hazard identification and initial risk ranking.
 - FTA (Fault Tree Analysis): A top-down technique for analyzing combinations of faults leading to a critical event.
 - V&V (Verification and Validation): Verification checks whether the design meets requirements; validation checks whether it fulfills the intended use.
 - ICD (Interface Control Document): A controlled document defining technical interfaces between systems or subsystems.
 - RACI: A responsibility matrix showing who is Responsible, Accountable, Consulted, and Informed.
 - GNC (Guidance, Navigation, and Control): The functions used to estimate state, guide motion, and control the vehicle.
 - AI: Artificial Intelligence, here limited to safety-governed sensing, estimation, and decision-support functions.
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